

MODULAR FORMS MMATH II YEAR

12 SEPTEMBER 2025
11:00 AM - 1:00 PM
MID-TERM EXAM 2025
MAX MARKS: 80
ANSWER ALL QUESTIONS

You are allowed one page A4 size cheat sheet

1. (a) Let $\Gamma = PSL(2, \mathbb{Z})$. For any $z \in \mathbb{H}$ define

$$\Gamma_z = \{\gamma \in \Gamma : \gamma z = z\}.$$

Determine: (i) Γ_i (ii) Γ_{1+2i} (iii) Γ_ρ where $\rho = e^{2\pi i/3}$. (3+3+4 marks)

(b) Determine if

$$\{z = x + iy : y > 9/10, |x| < 1/2\}$$

is a fundamental domain for Γ . (6 marks)

2. For $z \in \mathbb{H}$, let $\Delta(z) = g_2(z)^3 - 27g_3(z)^2$ be the discriminant form, and

$$j(z) = 1728 \frac{g_2(z)^3}{\Delta(z)}$$

be the j -invariant.

(a) Write down the valence formula (without proof) for modular forms of weight $2k$.

(b) Apply the valence formula to show that $\Delta(z) \neq 0$ for $z \in \mathbb{H}$.

(c) Using valence formula show that $j : \Gamma \backslash \mathbb{H} \rightarrow \mathbb{C}$ defines a bijection. (4+5+15 marks)

3. (a) Write down (without proof) the Petersson trace formula for the space $S_{2k}(\Gamma)$.

(b) Let $\tau(n)$ be the Ramanujan τ -function. Using (a) show that there exists an absolute constant $\gamma \neq 0$ such that

$$\frac{\gamma |\tau(n)|^2}{n^{11}} = 1 + 2\pi \sum_{c=1}^{\infty} \frac{S(n, n; c)}{c} J_{11} \left(\frac{4\pi n}{c} \right)$$

where $S(n, n; c)$ is the Kloosterman sum and $J_{11}(x)$ is the J -Bessel function of order 11.

(c) Assuming the bounds $|S(n, n; c)| \ll c^\theta$ (with $\theta > 0$) and $|J_{11}(x)| \ll \min\{1, x^{11}\}$, show that

$$|\tau(n)| \ll n^{(11+\theta)/2}.$$

(6+9+10 marks)

4. Let $\tau(n)$ denote the Ramanujan τ function. Consider the Dirichlet series

$$D(s) = \sum_{n=1}^{\infty} \frac{\tau(n)}{n^s}.$$

(a) Write down the Ramanujan conjectures for $\tau(n)$.

(b) Assuming the Ramanujan conjectures, show that $D(s)$ converges absolutely in the half-plane $\operatorname{Re}(s) > 13/2$. In this region, show that

$$D(s) = \prod_{p \text{ prime}} \left(1 - \frac{\tau(p)}{p^s} + \frac{p^{11}}{p^{2s}} \right)^{-1}$$

(6+6+8=20 marks)